

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1746

Tunica 138 kV
March 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1746. The Interconnection Customer proposes a 56.5 MW solar photovoltaic facility interconnecting at the Tunica 138 kV substation in the Maryland area. The request was received on May 14, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 644 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,938,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Clark County 345/138 kV Auto	Clark County-Artesia Country Club 345 kV	143.5	512	6.5
Air Force Base 345-138 kV Transformer Bank	Air Force Base-Tiger Bay 345 kV	134.4	541	5.4
Montano-Celeron 138 kV Line	Rio Dell-Celeron 138 kV	132.5	377	5.7
Rio Dell-Celeron 138 kV	Montano-Rio Dell 138 kV line	132.1	361	5.5
Rio Dell-Montano 138 kV	Rio Dell-Celeron 138 kV L	132.1	381	5.7
North Mesa-Sandalfot 138 kV Line	Celeron Substation-Yelm Pp 138 kV line	126.3	355	5.2
Columbus City 138/1kV Auto	International Paper Valliant Ok-Kearneyville 345 kV	125.9	495	6.0
Sandalfot Substation-Celeron 138 kV	Yelm Pp Substation-Celeron 138 kV line	125.7	369	5.2
Killingsworth Substation-Yelm Pp 138 kV	Celeron-Sandalfot 138 kV	125.3	365	5.1
Celeron-Yelm Pp 138 kV	Celeron-Sandalfot 138 kV Line	124.8	376	5.1
Healdsburg-Sintek 138 kV	Urban-American 345 kV	113.1	237	6.5
Healdsburg-W Batesville Substation 138 kV	Urban-American 345 kV	112.9	226	6.5
Islamorada-Prospect Central Substation 138 kV L	American-Urban 345 kV Line	111.3	294	5.3
Bedington-W Batesville 138 kV Line	Urban-American 345 kV Line	111.1	222	6.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
St. John-Rio Dell 138 kV	Kearneyville-International Paper Valliant Ok 345 kV	111.1	634	7.4
Lascassas-Bedington 138 kV Line	American-Urban 345 kV	110.1	211	6.5
Edens Ridge-Midset 138 kV	Urban-American 345 kV L	109.3	333	5.9
Columbus City 345/1kV Autotransformer	Kearneyville-International Paper Valliant Ok 345 kV	107.9	563	6.0
Islamorada-St. John 138 kV	Urban-American 345 kV	107.8	299	5.3
Marvin Griffin Rd-Prospect Central 138 kV L	American-Urban 345 kV	104.5	314	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clark County 345/138 kV Auto	Replace transformer with higher-rated unit	\$14,380,000	4% (\$523,000)
Air Force Base 345/138 kV Auto	Replace transformer with higher-rated unit	\$16,802,000	5% (\$813,000)
Montano-Celeron 138 kV	Rebuild line with higher-capacity conductor	\$1,393,000	51% (\$709,000)
Celeron-Rio Dell 138 kV Line	Rebuild line with higher-capacity conductor	\$5,531,000	51% (\$2,815,000)
Rio Dell-Montano 138 kV	Rebuild line with higher-capacity conductor	\$2,415,000	51% (\$1,229,000)
North Mesa-Sandalfoot 138 kV Line	Rebuild line with higher-capacity conductor	\$1,747,000	44% (\$768,000)
Columbus Cty 138-1 kV TX	Replace transformer with higher-rated unit	\$3,882,000	8% (\$322,000)
Sandalfoot-Celeron Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,944,000	44% (\$1,734,000)
Yelm Pp-Killingsworth 138 kV	Rebuild line with higher-capacity conductor	\$1,263,000	37% (\$473,000)
Celeron-Yelm Pp 138 kV	Rebuild line with higher-capacity conductor	\$7,306,000	37% (\$2,734,000)
Healdsburg-Sintek 138 kV	Rebuild line with higher-capacity conductor	\$2,434,000	34% (\$820,000)
Healdsburg-W Batesville Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,561,000	34% (\$526,000)
Islamorada-Prospect Central 138 kV	Rebuild line with higher-capacity conductor	\$4,789,000	100% (\$4,789,000)
Bedington-W Batesville 138 kV Line	Rebuild line with higher-capacity conductor	\$2,374,000	34% (\$800,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Dell-St. John 138 kV	Rebuild line with higher-capacity conductor	\$6,736,000	53% (\$3,567,000)
Lascassas-Bedington 138 kV Line	Rebuild line with higher-capacity conductor	\$10,730,000	34% (\$3,616,000)
Midset-Edens Ridge 138 kV	Reconductor line	\$2,822,000	7% (\$185,000)
Columbus City 345-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$11,080,000	8% (\$918,000)
Islamorada-St. John 138 kV	Reconductor line	\$2,097,000	100% (\$2,097,000)
Marvin Giffin Road-Prospect Central 138 kV	Reconductor line	\$1,500,000	100% (\$1,500,000)
	Total allocated to project		\$30,938,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.